

VICTOR IZEKOR

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EDUCATION

University of Maryland, Baltimore County (UMBC)

Baltimore, MD

Bachelor of Science in Bioinformatics — Computer Science & Engineering Track

Expected May 2027

Relevant Coursework: Computer Science I & II, Data Structures & Algorithm Analysis (CMSC 341), Object-Oriented Programming, Statistics & Quantitative Analysis, Bioinformatics Algorithms

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript, HTML/CSS — primary proficiency in C++ and Python

Data Structures & Algorithms: Hash tables, dynamic programming, object-oriented design, algorithm complexity analysis (Big-O), debugging & testing frameworks

Software Practices: Agile methodology, requirements definition, modular architecture, design documentation, version control (Git)

Tools: Microsoft Office Suite, GCC/G++ compiler, command-line development environments

EXPERIENCE

Johns Hopkins Bayview Medical Center

Baltimore, MD

Patient Care Technician Intern — Neuro ICU

October 2025 – Present

- Work within a cross-functional clinical team, communicating precise, time-critical information under pressure — collaboration and communication skills that translate directly to fast-moving engineering teams.
- Interpret and document complex patient data accurately for multiple stakeholders, reflecting the same attention to detail and clear technical writing expected in software development.
- Operate reliably in a high-stakes, deadline-driven environment, a habit that carries directly into shipping software under real constraints.

PROJECTS

Custom Hash Table — Data Structures & Algorithm Design

UMBC CMSC 341

- Engineered a hash table in C++ using separate chaining, with dynamic resizing, collision resolution, and full manual memory management.
- Validated correctness with a dedicated testing framework across edge cases; performed Big-O analysis and tuned for $O(1)$ average-case lookup.

Cipher Encryption/Decryption Engine — Applied Software Design

UMBC CMSC 202

- Implemented three encryption algorithms (Atbash, Scytale, Ong) in C++ using a modular class hierarchy with clear interface contracts.
- Applied encapsulation, inheritance, and polymorphism to build a system that supports new cipher modules without refactoring core logic.
- Enforced rigorous input validation and output correctness across all test cases.

Pulsify — Full-Stack Web Application

February 2025

Howard University Bison Bytes Hackathon

- Designed and built a full-stack web application (Python/Flask backend, JavaScript/HTML/CSS frontend) end-to-end within 24 hours.
- Owned requirements definition, system architecture, and key implementation decisions within an interdisciplinary team, delivering a working prototype from concept to demo.

Sequence Alignment Algorithms — Applied Algorithm Design

UMBC Bioinformatics Coursework

- Implemented Needleman-Wunsch and Smith-Waterman dynamic programming algorithms in Python, comparing runtime and accuracy trade-offs between global and local optimization strategies, and validated output against real structured datasets.

ADDITIONAL INFORMATION

Interests: Full-stack development, applied machine learning, algorithm design, healthcare technology, computational biology